
CS 212 : Design and Analysis of Algorithms

Course Code : CS 212

Course Name : Design and Analysis of Algorithms

L-T-P-C : 3-0-2-4

Intended for : BTech (Semester 3 or 4)

Prerequisite : IC253: Programming and Data Structures, or equivalent for UG students

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- Review of Data Structures from IC course: Asymptotic Notations and recurrence relations, Insertion Sort, Merge Sort, QuickSort and Randomization (5 H)
- Sorting Lower Bounds and Non-Comparison Sorting algorithms like Counting Sort and Radix Sort (2 H)
- Order Statistics: Minima and Maxima, Quickselect, Median of Medians (3 H)
- Applying sorting techniques to Computational Geometry : Closest pair problem, Convex Hull of points in a plane (3 H)
- Advanced topics in Sorting and Searching:- Binary Search Trees, Red-Black Trees, Augmentation of Binary Search Trees (8 H)
- Graph Algorithms: Basics of Graphs, BFS and DFS, Topological Sorting, MST(using Kruskal and Prim's), Union-Find Data structure, Single-Source Shortest Path Dijkstra's Algorithm and Bellman-Ford /Maximum Flow (Ford Fulkerson) (8 H)
- Dynamic Programming: Longest Common Subsequence, Matrix Multiplication (2 H)
- Amortized analysis: aggregate analysis, accounting, potential method (3 H)
- Advanced Data Structures: B-Trees, Binomial and Fibonacci/Quake Heaps (3 H)
- Computational complexity: Problem classes: P, NP, NP-complete, NP-hard. Reductions and Examples of NP-complete problems. (3 H)
- Dealing with NP-hard problems:- Approx/Randomized Algorithms, Parameterized Complexity (2 H)
- Coding Lab which covers topics discussed in this course (28 H)

Textbooks:

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, Introduction to Algorithms, MIT Press, 3/e, 2009.
2. J. Kleinberg and E. Tardos, Algorithm Design, Pearson, 2006.



References:

1. S. Dasgupta, C. H. Papadimitriou, U. V. Vazirani, Algorithms, McGraw-Hill, 2006.
2. S. S. Skiena, The Algorithm Design Manual, Springer, 2/e, 2008